Class-6 Chapter 5: Measurement of Length and Motion— Complete Solution NCERT New Syllabus

Page 75 – Stop and Reflect

A. Choose the correct answer

1. Which of the following is not a unit of length?

(a) Inch
(b) Metre
(c) Litre
(d) Feet

Answer: (c) Litre. Litre is a unit of capacity, not length.

2. The length between the tip of the second finger and the elbow is known as _____.

(a) Handspan
(b) Arm length
(c) Cubit
(d) Footstep

Answer: (b) Arm length.

3. Which of the following is non-periodic motion?

(a) Flapping of an insect's wings
(b) Movement of a swing
(c) Movement of a pendulum in a clock
(d) Beating of heart

Answer: (a) Flapping of an insect's wings. The other examples are commonly treated as periodic motions in this chapter.

4. Which of the following is a standard unit of measuring length?

(a) Cubit
(b) Arm length

(c) Metre
(d) Footstep

Answer: (c) Metre.

5. Which of the following is incorrect about linear motion?

(a) It is a type of translational motion.
(b) A car moving on a straight road shows linear motion.
(c) A truck moving on a curved road shows linear motion.
(d) None of these.

Answer: (c) A truck moving on a curved road shows linear motion. A truck moving on a curved road shows curvilinear motion.

B. Write True or False. Also, correct the incorrect statements.

1. An object is said to be in motion when its position changes with time.

Answer: True.

2. Non-standard units of measurement are reliable and precise.

Answer: False. Non-standard units are not always reliable or precise because they may differ from person to person.

3. The length of a curved line can be measured using a ruler.

Answer: False. The length of a curved line can be measured using a thread and then measuring the thread with a ruler.

4. Ships and boats are examples of water transport.

Answer: True.

5. A person taking a turn on a curved path is said to be in circular motion.

Answer: False. A person moving along a curved path shows curvilinear motion. Circular motion occurs when an object moves along a circular path.

C. Match the columns

Column A	Correct Match
1. SI unit of length	(e) Metre (m)
2. Distance between shoulder and tip of second finger	(c) Arm length
3. Used to measure length of an object	(b) Ruler
4. Motion of a train on a curved track	(a) Curvilinear
5. Revolution of the moon around the earth	(d) Rotational

D. Interconvert the units

1. 1500 cm = _____ m

Answer: 1500 ÷ 100 = 15 m

2. 2 m = _____ mm

Answer: 2 × 1000 = 2000 mm

3. 1000 km = _____ m

Answer: 1000 × 1000 = 10,00,000 m

4. 10 dm = _____ cm

Answer: 10 × 10 = 100 cm

5. 10 m = _____ km

Answer: 10 ÷ 1000 = 0.01 km

E. Identify the type of motion

1. Tram moving on a straight track

Answer: Linear motion.

2. A lift going downwards

Answer: Linear motion.

3. An ant moving on the floor

Answer: Random motion, if it moves in an irregular direction.

4. The motion of a stone tied to a rope

Answer: Circular motion.

5. The movement of a spring after compression

Answer: Oscillatory motion.

Page 76 – Crisp and Accurate

A. Define the following terms

1. Measurement

Answer: Measurement is the process of finding the size or quantity of something by comparing it with a standard unit.

2. Unit

Answer: A unit is a fixed standard quantity used to measure something. For example, metre is a unit of length.

3. Motion

Answer: An object is said to be in motion when its position changes with time in relation to a reference point.

4. Periodic Motion

Answer: Periodic motion is a type of motion that repeats itself at regular intervals of time. For example, the motion of a pendulum.

5. Non-periodic Motion

Answer: Non-periodic motion is a type of motion that does not repeat itself at regular intervals of time. For example, the irregular movement of an insect.

B. Answer the following questions briefly

1. Name the non-standard units of measurement.

Answer: Some common non-standard units of measurement are:

- Handspan
- Cubit
- Footstep
- Arm length

2. What do you mean by International System of Units?

Answer: The International System of Units (SI) is a system of standard units used all over the world for measurement. For example, metre is the SI unit of length, kilogram is the SI unit of mass and second is the SI unit of time.

3. What is the length of your science textbook in terms of your handspan?

Answer: Place your handspan repeatedly along the length of your science textbook and count the number of handspans. The answer will depend on the size of the book and your hand.

Note: Students should measure their own textbooks to find the exact answer.

4. When is the motion of an object said to be random?

Answer: The motion of an object is called random motion when it moves in an irregular direction without following a fixed path. For example, the movement of a flying butterfly.

5. What is oscillatory motion?

Answer: Oscillatory motion is the repeated to-and-fro movement of an object about its mean position. For example, the movement of a swing.

Think and Explain – Long Answer Questions

1. What is the need for measurement? Explain.

Answer: Measurement is important in our daily life because it helps us find the exact size, length, weight and quantity of different things.

The main needs for measurement are:

1. To know the exact size: Measurement helps us find the length, height and width of objects.
2. In buying and selling: We use measurement to measure the weight and quantity of goods.
3. In construction: Builders use measurement to construct houses, roads and bridges.

4. In science: Scientists use standard units to perform experiments and record results.
5. For comparison: Measurement helps us compare the sizes and quantities of different objects.

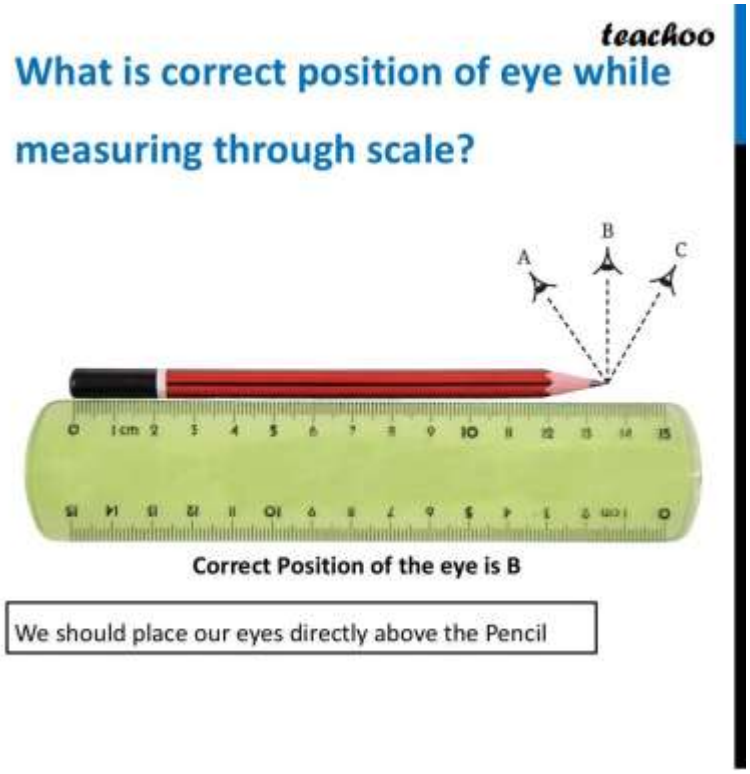
For example, a tailor measures our body before stitching clothes, and a shopkeeper measures vegetables before selling them.

2. Describe the steps for measuring the length of an object with the help of a ruler.

Answer: We can measure the length of an object using a ruler by following these steps:

1. Take a ruler with centimetre and millimetre markings.
2. Place the ruler along the length of the object.
3. Keep the zero mark of the ruler at one end of the object.
4. Look at the reading at the other end of the object.
5. Read the measurement carefully, keeping your eye directly above the mark to avoid parallax error.
6. Write the measurement with its proper unit, such as centimetre (cm).

Example: If an object starts at 0 cm and ends at 8 cm, its length is 8 cm.



3. Explain how the length of a curved line can be measured with the help of a diagram.

Answer: A curved line cannot be measured directly with a straight ruler. We can measure it using a thread and a ruler.

Materials Required:

- A piece of thread
- A ruler
- A pencil

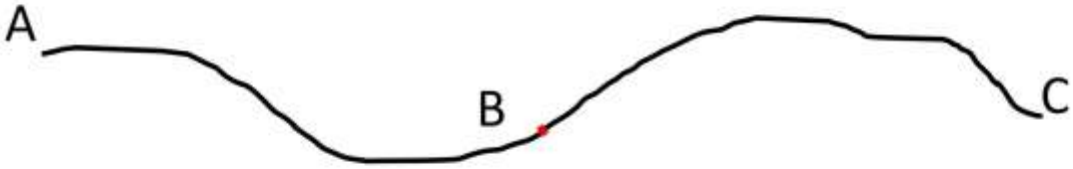
Procedure:

1. Take a piece of thread.
2. Place one end of the thread at the starting point of the curved line.
3. Carefully place the thread along the entire curved line.
4. Mark the ending point on the thread.
5. Straighten the thread without stretching it.
6. Measure the marked length using a ruler.

Observation: The length of the straightened thread gives the length of the curved line.

Conclusion: The length of a curved line can be measured with the help of a thread and a ruler.

Using string to measure circumference



Remember, use the same tip of the string when measuring using the ruler.

If the point is at AB (B is the marked point), measure the length AB with the ruler and not BC.

istinguish between periodic motion and non-periodic motion.

Answer: Periodic motion repeats itself at regular intervals of time, whereas non-periodic motion does not repeat itself at regular intervals.

Periodic Motion	Non-periodic Motion
It repeats at regular intervals of time.	It does not repeat at regular intervals of time.
It follows a repeated pattern.	It does not follow a regular repeated pattern.
It is generally predictable.	It is generally less predictable.
Example: Motion of a pendulum.	Example: Irregular movement of an insect.
Example: Beating of the heart.	Example: Movement of a person in a crowded place.

Examples of periodic motion:

- Movement of a swing.
- Movement of a pendulum.
- Beating of the heart.

Examples of non-periodic motion:

- Movement of a flying butterfly.
- Movement of a person in a crowded place.
- Irregular movement of an insect.

5. Describe motion of more than one type by taking four examples.

Answer: Some objects show more than one type of motion at the same time. Four examples are:

1. A rolling bicycle wheel

- The wheel rotates around its axle.
- At the same time, the bicycle moves forward.
- Therefore, it shows rotational and translational motion.

2. The Earth

- The Earth rotates on its own axis.
- It also revolves around the Sun.
- Therefore, it shows rotational and periodic motion.

3. A moving ceiling fan

- The blades rotate around a fixed axis.
- The rotation repeats regularly.
- Therefore, it shows rotational and periodic motion.

4. A child sitting on a moving swing

- The swing moves to and fro.
- This movement repeats after regular intervals.
- Therefore, it shows oscillatory and periodic motion.



(a) Motion of a ceiling fan

(b) Motion of Earth about its axis

(c) Motion of a spinning top

Examples of rotational motion

Quick Revision – Important Points

1. Measurement is the process of comparing an object with a standard unit.
2. Metre (m) is the SI unit of length.
3. A ruler is used to measure the length of small objects.
4. Handspan, cubit and footstep are non-standard units.
5. One metre = 100 centimetres.
6. One metre = 1000 millimetres.
7. One kilometre = 1000 metres.
8. Motion occurs when an object's position changes with time.
9. Linear motion takes place along a straight line.
10. Curvilinear motion takes place along a curved path.
11. Circular motion takes place along a circular path.
12. Rotational motion occurs when an object rotates about an axis.
13. Oscillatory motion is repeated to-and-fro movement.
14. Periodic motion repeats at regular intervals.
15. Non-periodic motion does not repeat at regular intervals.
16. Random motion takes place in an irregular manner.
17. A thread can be used to measure the length of a curved line.

Remember: Always use standard units and proper measuring instruments to get accurate measurements.